Updating Ubuntu

```
afsar-rasool@afsar-rasool:~/Desktop$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://pk.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [
673 kB]
```

Installing kernel development dependencies.

```
afsar-rasool@afsar-rasool:~/Desktop$ sudo apt install build-essential libncurses
-dev bison flex libssl-dev libelf-dev bc dwarves wget git fakeroot
[sudo: authenticate] Password:
bc is already the newest version (1.07.1-4build1).
bc set to manually installed.
wget is already the newest version (1.25.0-2ubuntu4.4).
wget set to manually installed.
Installing:
  bison           dwarves   flex  libelf-dev      libssl-dev
  build-essential fakeroot  git   libncurses-dev

Installing dependencies:
  bzip2                     libalgorithm-merge-perl  libitm1
  dpkg-dev                  libasan8                 liblsan0
  g++                       libcc1-0                 libquadmath0
  g++-15                    libcrypt-dev             libstdc++-15-dev
```

Kernel After Update:a

```
afsar-rasool@afsar-rasool:~/Desktop$ uname -r
7.0.0-34-generic
```

```
afsar-rasool@afsar-rasool:~/Desktop$ ls /boot | grep vmlinuz
vmlinuz
vmlinuz-7.0.0-34-generic
vmlinuz.old
```

GCC & Make:

```
afsar-rasool@afsar-rasool:~/Desktop$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
afsar-rasool@afsar-rasool:~/Desktop$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
```

Creating Kernel Workspace:

```
afsar-rasool@afsar-rasool:~$ mkdir kernels
afsar-rasool@afsar-rasool:~$ cd kernels
afsar-rasool@afsar-rasool:~/kernels$
```

Downloading and Extracting Linux Kernel 7.2.7

```
afsar-rasool@afsar-rasool:~/kernels$ wget https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.2.7.tar.x
z
--2026-09-25 15:15:38--  https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.2.7.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 151.101.193.176, 151.101.1.176, 151.101.65.176, ...
Connecting to cdn.kernel.org (cdn.kernel.org)|151.101.193.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 160321592 (153M) [application/x-xz]
Saving to: 'linux-7.2.7.tar.xz'

linux-7.2.7.tar.xz        100%[===================================>] 152.89M  1.43MB/s    in 2m 11s

2026-09-25 15:17:51 (1.17 MB/s) - 'linux-7.2.7.tar.xz' saved [160321592/160321592]
```

```
afsar-rasool@afsar-rasool:~/kernels$ ls -lh
total 153M
-rw-rw-r-- 1 afsar-rasool afsar-rasool 153M Sep 21 18:54 linux-7.2.7.tar.xz
```

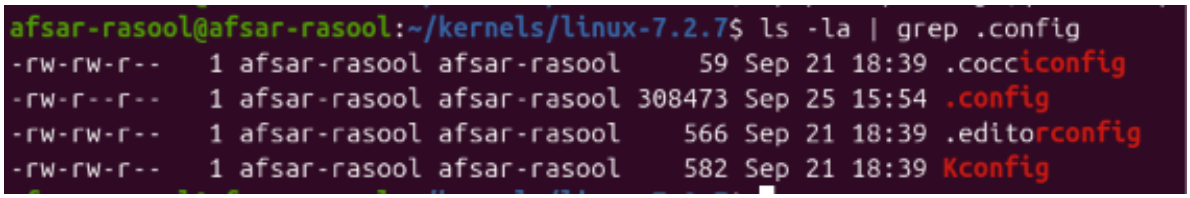
```
afsar-rasool@afsar-rasool:~/kernels$ tar -xf linux-7.2.7.tar.xz
afsar-rasool@afsar-rasool:~/kernels$ ls
linux-7.2.7  linux-7.2.7.tar.xz
afsar-rasool@afsar-rasool:~/kernels$ cd linux-7.2.7
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ ls
COPYING        Kbuild    MAINTAINERS  arch   crypto   include   ipc     mm    samples   sound  virt
CREDITS        Kconfig   Makefile     block  drivers  init      kernel  net   scripts   tools
Documentation  LICENSES  README       certs  fs       io_uring  lib     rust  security  usr
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$
```

```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ uname -r
7.0.0-34-generic
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ ls /boot | grep config
config-7.0.0-34-generic
```

Copy current running kernel config into the new source tree:

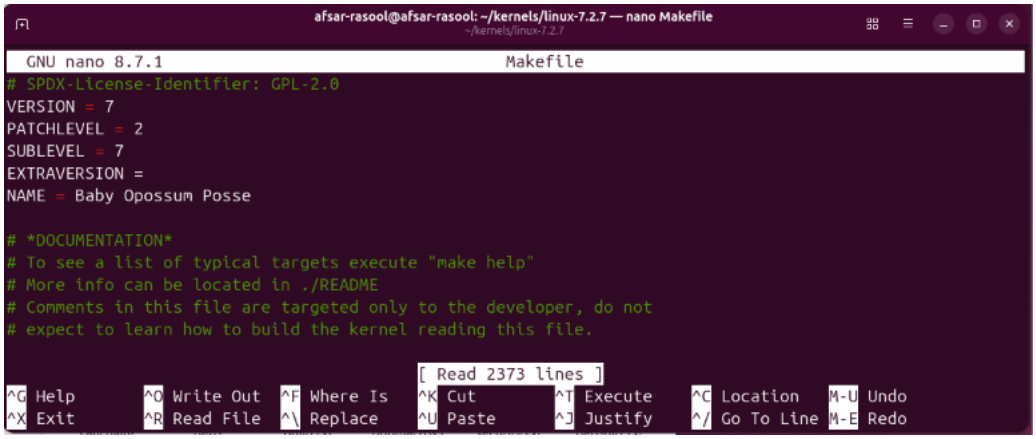
```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ cp /boot/config-$(uname -r) .config
```

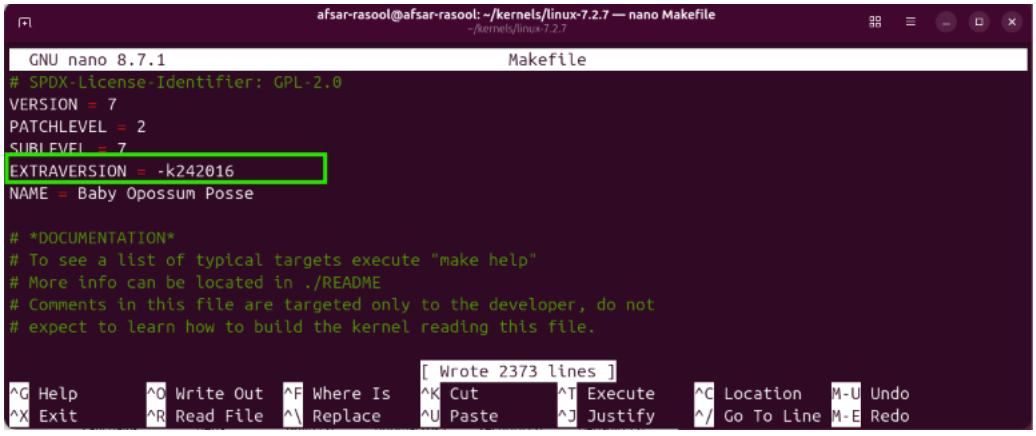
Verify Config File:

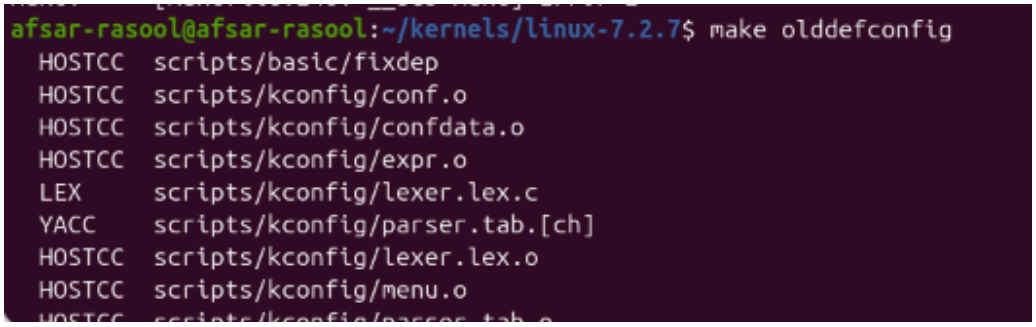


Renaming New Kernel with Roll Number (k2420xx):





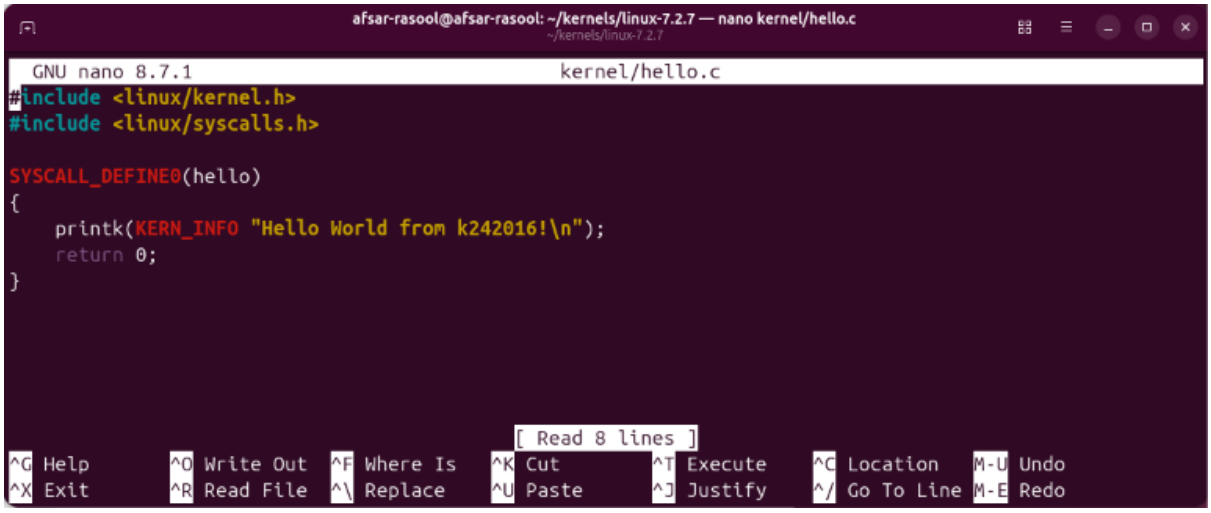




```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ make kernelrelease
7.2.7-k242016
```

Adding Hello World Syscall:

```
afsar-rasool:~/kernels/linux-7.2.7$ nano kernel/hello.c
```



SYSCALL_DEFINE0(hello) creates a system call named hello with **zero arguments**.

The Linux documentation recommends using the SYSCALL_DEFINEn() macros when implementing system calls. Linux Kernel Documentation

printk() writes into the kernel log.

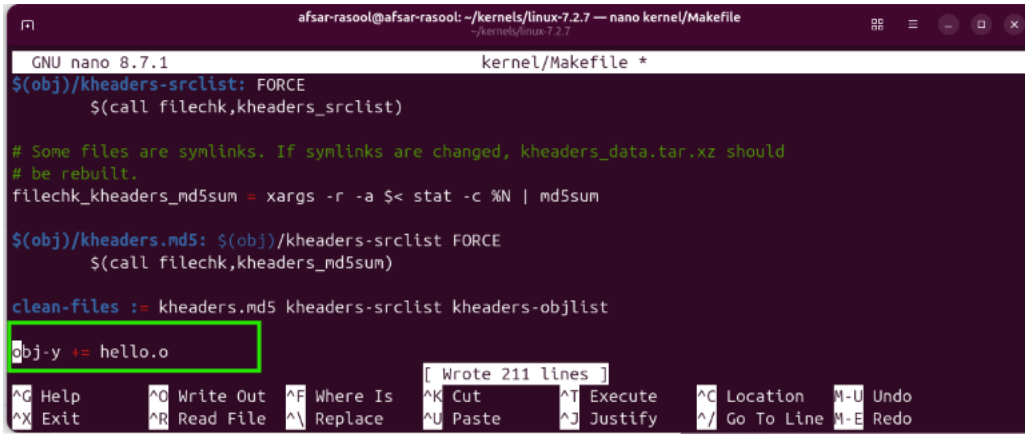
Later:

dmsg

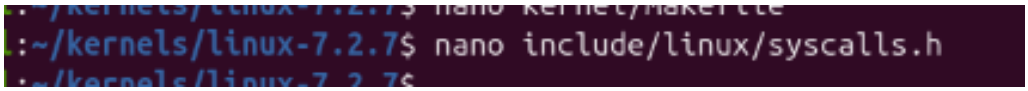
to see kernel logs

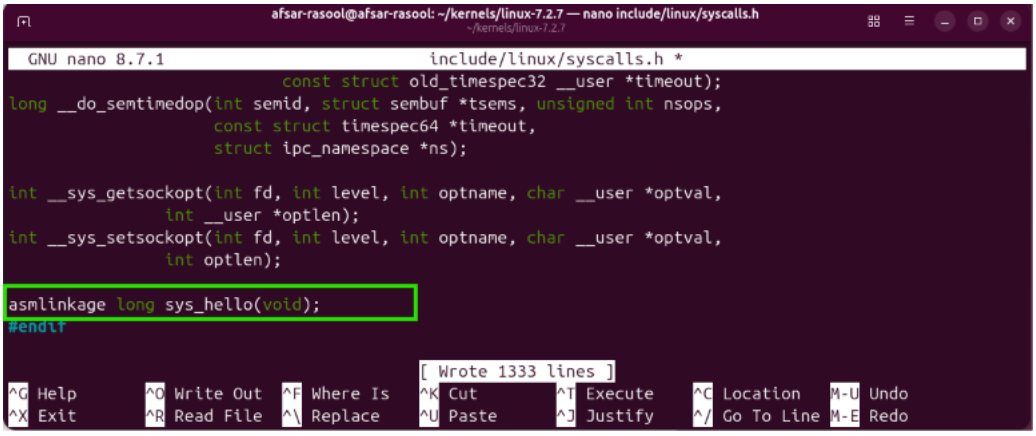
Telling the **kernel build system to compile** hello.c

```
$ nano kernel/Makefile
```



Add the syscall prototype:



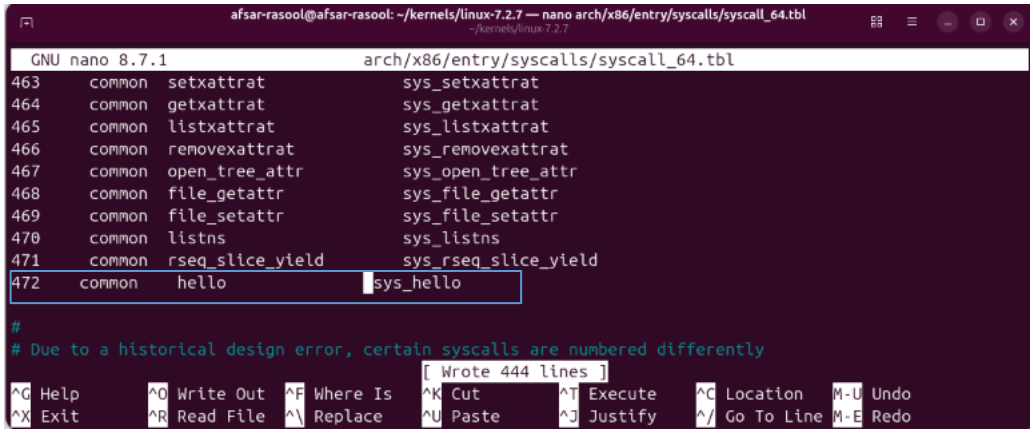


SystemCall Table:

- ⇨ The syscall table connects a **syscall number** to your kernel function:

472
↓
hello
↓
sys_hello
↓
SYSCALL_DEFINE0(hello)

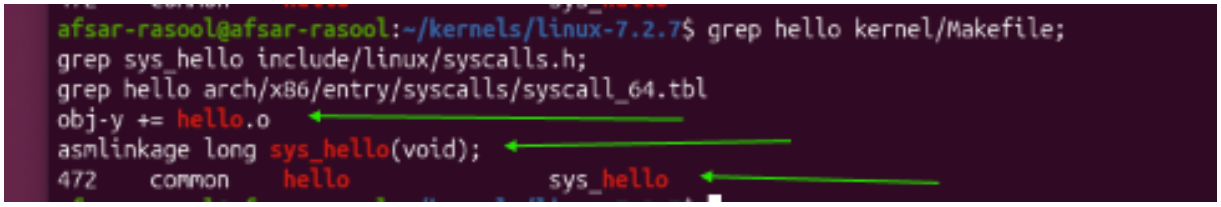
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ nano arch/x86/entry/syscalls/syscall_64.tbl



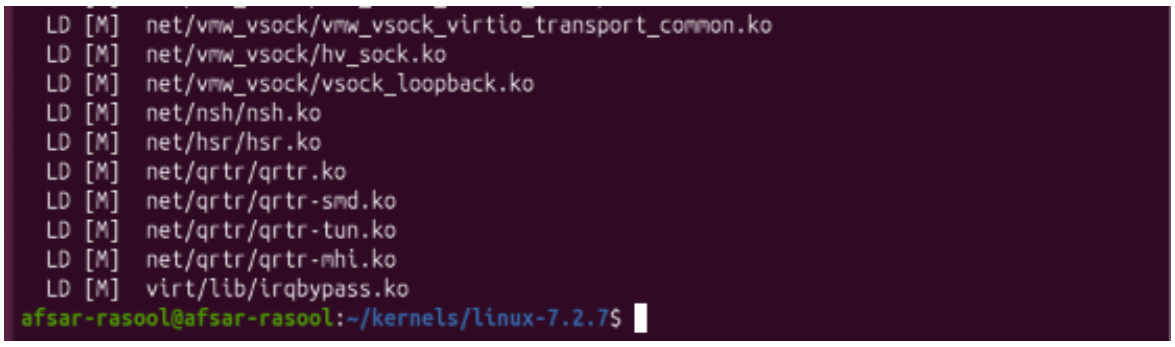
grep hello kernel/Makefile;

grep sys_hello include/linux/syscalls.h;

grep hello arch/x86/entry/syscalls/syscall_64.tbl



Compilation Complete:



Installing kernel Modules:

```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ sudo make modules_install
[sudo: authenticate] Password:
  SYMLINK /lib/modules/7.2.7-k242016/build
  INSTALL /lib/modules/7.2.7-k242016/modules.order
  INSTALL /lib/modules/7.2.7-k242016/modules.builtin
  INSTALL /lib/modules/7.2.7-k242016/modules.builtin.modinfo
  INSTALL /lib/modules/7.2.7-k242016/modules.builtin.ranges
  INSTALL /lib/modules/7.2.7-k242016/kernel/arch/x86/events/amd/amd-uncore.ko
  SIGN    /lib/modules/7.2.7-k242016/kernel/arch/x86/events/amd/amd-uncore.ko
  INSTALL /lib/modules/7.2.7-k242016/kernel/arch/x86/events/intel/intel-cstate.ko
  SIGN    /lib/modules/7.2.7-k242016/kernel/arch/x86/events/intel/intel-cstate.ko
  INSTALL /lib/modules/7.2.7-k242016/kernel/arch/x86/events/rapl.ko
  SIGN    /lib/modules/7.2.7-k242016/kernel/arch/x86/events/rapl.ko
  INSTALL /lib/modules/7.2.7-k242016/kernel/arch/x86/kernel/cpu/mce/mce-inject.ko
  SIGN    /lib/modules/7.2.7-k242016/kernel/arch/x86/kernel/cpu/mce/mce-inject.ko
  INSTALL /lib/modules/7.2.7-k242016/kernel/arch/x86/kernel/msr.ko
  SIGN    /lib/modules/7.2.7-k242016/kernel/arch/x86/kernel/msr.ko
```

Installing Kernel:

```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ sudo make install
  INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/dracut 7.2.7-k242016 /boot/vmlinuz-7.2.7-k242016
dracut: Generating /boot/initrd.img-7.2.7-k242016
run-parts: executing /etc/kernel/postinst.d/kdump-tools 7.2.7-k242016 /boot/vmlinuz-7.2.7-k2
42016
kdump-tools: Generating /var/lib/kdump/initrd.img-7.2.7-k242016
run-parts: executing /usr/share/kernel/postinst.d/touch-reboot-required 7.2.7-k242016 /boot/
vmlinuz-7.2.7-k242016
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 7.2.7-k242016 /boot/vmlinuz-
7.2.7-k242016
run-parts: executing /etc/kernel/postinst.d/update-notifier 7.2.7-k242016 /boot/vmlinuz-7.2.
7-k242016
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 7.2.7-k242016 /boot/vmlin
uz-7.2.7-k242016
I: /boot/initrd.img is now a symlink to initrd.img-7.2.7-k242016
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 7.2.7-k242016 /boot/vmlinuz-7.2.7
-k242016
Sourcing file `/etc/default/grub'
```

```
Found memtest86+ 64bit EFI image: /boot/mt86+x64
Found memtest86+ 32bit EFI image: /boot/mt86+ia32
Found memtest86+ 64bit image: /boot/mt86+x64
Found memtest86+ 32bit image: /boot/mt86+ia32
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
```

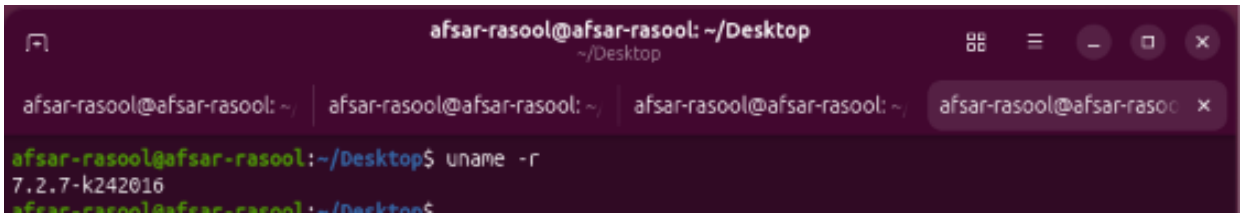
Updating GRUB:

```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ sudo update-grub
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/kdump-tools.cfg'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.2.7-k242016
Found initrd image: /boot/initrd.img-7.2.7-k242016
Found linux image: /boot/vmlinuz-7.0.0-34-generic
Found initrd image: /boot/initrd.img-7.0.0-34-generic
Found memtest86+ 64bit EFI image: /boot/mt86+x64
Found memtest86+ 32bit EFI image: /boot/mt86+ia32
Found memtest86+ 64bit image: /boot/mt86+x64
Found memtest86+ 32bit image: /boot/mt86+ia32
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
```

Verifying bootable files:

```
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ ls /boot | grep 7.2.7
System.map-7.2.7-k242016
config-7.2.7-k242016
initrd.img-7.2.7-k242016
vmlinuz-7.2.7-k242016
afsar-rasool@afsar-rasool:~/kernels/linux-7.2.7$ ls /lib/modules | grep 7.2.7
7.2.7-k242016
```

My Kernel Running:



System Call Execution:

Creating a program that make a system call:



Compilation & Execution:



Verifying Kernel Logs:

